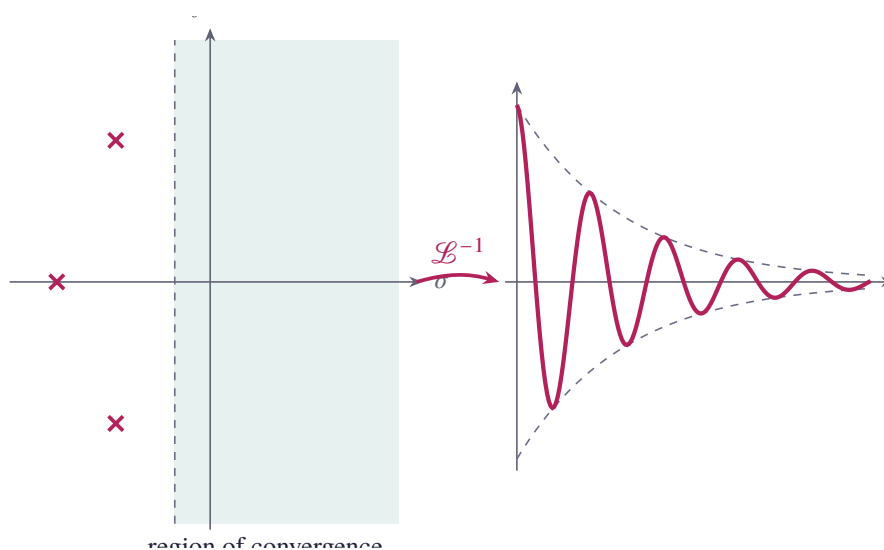


The Laplace Transform

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7.1 The One-Sided Laplace Transform

 **Definition 7.1** | The one-sided Laplace transform

$$F(s) = \mathcal{L}\{f(t)\} \triangleq \int_0^{\infty} f(t) e^{-st} dt \quad \Longleftrightarrow \quad f(t) = \mathcal{L}^{-1}\{F(s)\}$$

The Laplace transform is one of the important tools in various engineering and mathematical applications. One of its applications is solving differential equations (read Chapter 6 of the course textbook).

The Laplace transform is a linear operator:

$$\mathcal{L}\{af(t) + bg(t)\} = a \mathcal{L}\{f(t)\} + b \mathcal{L}\{g(t)\}$$

 **Example 7.2** | Example 1

$$f(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \Rightarrow \mathcal{L}\{f(t)\} = F(s) = ?$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} 1 \times e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

 **Example 7.3** | Example 2

$$f(t) = \begin{cases} e^{at} & t \geq 0 \\ 0 & t < 0 \end{cases} \Rightarrow \mathcal{L}\{f(t)\} = F(s) = ?$$

$$\mathcal{L}\{f(t)\} = \int_0^{\infty} e^{at} \times e^{-st} dt = \int_0^{\infty} e^{-(s-a)t} dt = -\frac{1}{s-a} e^{-(s-a)t} \Big|_0^{\infty} = \frac{1}{s-a}$$

 **Example 7.4** | Example 3

$$\mathcal{L}\{\cosh(at)\} = \mathcal{L}\left\{\frac{1}{2}e^{at} + \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} + \frac{1}{2}\mathcal{L}\{e^{-at}\} = \frac{1}{2}\frac{1}{s-a} + \frac{1}{2}\frac{1}{s+a} = \frac{s}{s^2 - a^2}$$

 **Example 7.5** | Example 4

$$\mathcal{L}\{\sinh(at)\} = \mathcal{L}\left\{\frac{1}{2}e^{at} - \frac{1}{2}e^{-at}\right\} = \frac{1}{2}\mathcal{L}\{e^{at}\} - \frac{1}{2}\mathcal{L}\{e^{-at}\} = \frac{1}{2}\frac{1}{s-a} - \frac{1}{2}\frac{1}{s+a} = \frac{a}{s^2 - a^2}$$

 **Example 7.6** | Example 5

Using $\int e^{ax} \cos nx dx = \frac{e^{ax}}{a^2 + n^2} (a \cos nx + n \sin nx)$:

$$\mathcal{L}\{\cos \omega t\} = \int_0^{\infty} \cos \omega t e^{-st} dt = \frac{e^{-st}}{s^2 + \omega^2} (-s \cos \omega t + \omega \sin \omega t) \Big|_0^{\infty} = \frac{s}{s^2 + \omega^2}$$

Example 7.7 | Example 6

Using $\int e^{ax} \sin nx \, dx = \frac{e^{ax}}{a^2 + n^2} (a \sin nx - n \cos nx)$:

$$\mathcal{L}\{\sin \omega t\} = \int_0^\infty \sin \omega t \, e^{-st} \, dt = \frac{e^{-st}}{s^2 + \omega^2} (-s \sin \omega t - \omega \cos \omega t) \Big|_0^\infty = \frac{\omega}{s^2 + \omega^2}$$

Table of Laplace Transforms (Table 6.1 of the textbook; a longer table is in Section 6.9)

	$f(t)$	$\mathcal{L}(f)$		$f(t)$	$\mathcal{L}(f)$
1	1	$\frac{1}{s}$	7	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
2	t	$\frac{1}{s^2}$	8	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
3	t^2	$\frac{2!}{s^3}$	9	$\cosh at$	$\frac{s}{s^2 - a^2}$
4	t^n ($n = 0, 1, \dots$)	$\frac{n!}{s^{n+1}}$	10	$\sinh at$	$\frac{a}{s^2 - a^2}$
5	t^a (a positive)	$\frac{\Gamma(a+1)}{s^{a+1}}$	11	$e^{at} \cos \omega t$	$\frac{s-a}{(s-a)^2 + \omega^2}$
6	e^{at}	$\frac{1}{s-a}$	12	$e^{at} \sin \omega t$	$\frac{\omega}{(s-a)^2 + \omega^2}$

$$\Gamma(x) = \int_0^\infty e^{-t} t^{x-1} \, dt$$

Properties of the One-Sided Laplace Transform (the table of Section 6.8 of the textbook)

With the unit step function

$$u(t) \triangleq \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

Property	Formula
Definition of the Laplace transform	$F(s) = \mathcal{L}\{f(t)\} \triangleq \int_0^{\infty} f(t)e^{-st} dt$
Inverse Laplace transform	$f(t) = \mathcal{L}^{-1}\{F(s)\}$
Linearity	$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$
Time shift	$\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$
Shift in the s domain	$\mathcal{L}\{e^{at}f(t)\} = F(s-a)$
Time scaling	$\mathcal{L}\{f(at)\} = \frac{1}{a}F\left(\frac{s}{a}\right)$
Time derivative	$\mathcal{L}\{f^{(n)}(t)\} = s^n F(s) - s^{n-1}f(0) - s^{n-2}f^{(1)}(0) - \dots - f^{(n-1)}(0)$
Time integral	$\mathcal{L}\left\{\int_0^t f(\tau) d\tau\right\} = \frac{1}{s}F(s)$
Derivative in the s domain	$\mathcal{L}\{tf(t)\} = -F'(s)$
Integral in the s domain	$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^{\infty} F(v) dv$

7.2 Using the Laplace Transform to Solve Differential Equations

The steps for solving a differential equation with initial conditions using the Laplace transform:

1. convert the initial-value problem into an algebraic problem;
2. compute the solution of the algebraic problem;
3. solve the initial-value problem (return the solution to the time domain).



Figure 7.1 The steps of solving an initial-value problem with the Laplace transform: initial-value problem $\xrightarrow{1}$ algebraic problem $\xrightarrow{2}$ solution of the algebraic problem $\xrightarrow{3}$ solution of the initial-value problem

■ Ordinary Differential Equations

Example 7.8

Solve the differential equation $y'' - y = t$, $y(0) = 1$, $y'(0) = 1$ using the Laplace transform.

$$s^2 Y - sy(0) - y'(0) - Y = \frac{1}{s^2} \Rightarrow (s^2 - 1)Y(s) = s + 1 + \frac{1}{s^2}$$

$$Y(s) = \frac{s+1}{s^2-1} + \frac{1}{s^2(s^2-1)} = \frac{1}{s-1} + \left(\frac{1}{s^2-1} - \frac{1}{s^2} \right)$$

$$\begin{aligned} y(t) &= \mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2-1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} \\ &= e^t u(t) + \sinh t u(t) - t u(t) \end{aligned}$$

Partial Differential Equations

Example 7.9

Solve the following partial differential equation using the Laplace transform.

$$\frac{\partial w}{\partial x} + x \frac{\partial w}{\partial t} = 0, \quad w(x, 0) = 0, \quad w(0, t) = t u(t)$$

We take the Laplace transform of both sides with respect to t :

$$\mathcal{L}_t \left\{ \frac{\partial w}{\partial x} + x \frac{\partial w}{\partial t} \right\} = \mathcal{L}_t \left\{ \frac{\partial w}{\partial x} \right\} + x \mathcal{L}_t \left\{ \frac{\partial w}{\partial t} \right\} = 0$$

$$\frac{\partial}{\partial x} w(x, s) + x[s w(x, s) - w(x, 0)] = 0, \quad w(x, 0) = 0$$

$$\frac{\partial}{\partial x} w(x, s) + x s w(x, s) = 0 \Rightarrow \frac{dw}{w} = -s x dx \Rightarrow \ln w = -s \frac{x^2}{2} + \ln C \Rightarrow w(x, s) = C e^{-\frac{x^2}{2}s}$$

From $w(0, t) = t u(t)$ we have $w(0, s) = \mathcal{L}\{w(0, t)\} = \frac{1}{s^2} = C$:

$$w(x, s) = \frac{1}{s^2} e^{-\frac{x^2}{2}s}$$

Using the time-shift property $\mathcal{L}\{f(t-a)u(t-a)\} = e^{-as}F(s)$:

$$w(x, t) = \left(t - \frac{x^2}{2}\right) \times u\left(t - \frac{x^2}{2}\right)$$